# Lecture 5: Random Variable and Probability Distribution


Lecturer

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CLASS CODE

**TOAE203**

COURSE

**Statistics and Probability in Economics and Business**

# Agenda

**01**

### RANDOM VARIABLES

A function assigning a number to each outcome, and the distinction between discrete and continuous.

**02**

### EXPECTATION AND VARIANCE

The centre and the width of a distribution, applied to a choice between two investments.

**03**

### DISCRETE DISTRIBUTION

Bernoulli and binomial, stated as data-generating mechanisms rather than as formulas.

**04**

### CONTINUOUS DISTRIBUTION: FROM BARS TO AREA

Continuous variables, density, the cumulative function, and the normal distribution.

**05**

### SUMMARY

Key Takeaways, assessed output

RANDOM VARIABLES

# The proportion of firms above mean revenue

VN-LISTED, most recent year

—

**MEAN REVENUE, BILLION VND**

Computed from the current extract before the session.

—%

**FIRMS ABOVE THAT MEAN**

The proportion is governed by the shape of the distribution, not by the mean.

**What proportion of firms report revenue above the mean? (a) 50% · (b) 40% · (c) 30% · (d) 20%. Vote before the reveal.**

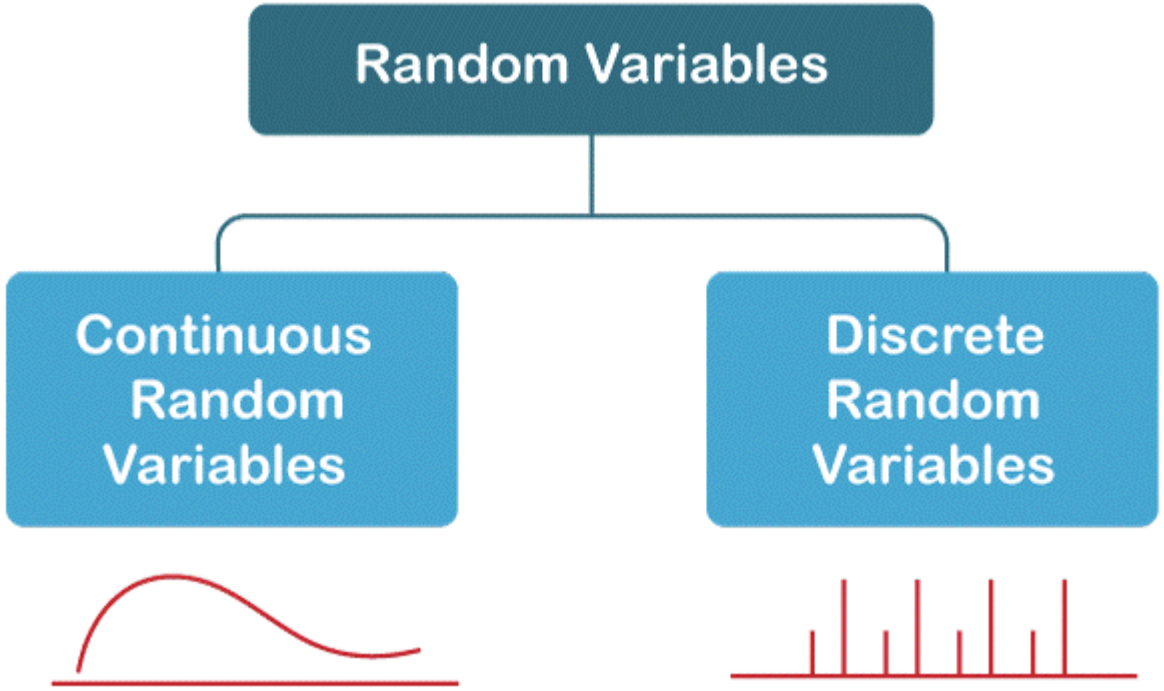
# Random variables: definition and classification

**DEFINITION**

## A rule that assigns a number to each outcome

A **random variable** (biến ngẫu nhiên) is a function mapping every outcome of an experiment to a number. It is the assignment rule, not the number itself.

- **Discrete**: the values can be listed one by one
- **Continuous**: between any two values a further value exists




**The key to decide a variable is random or not is the “UNCERTAINTY” of the results**

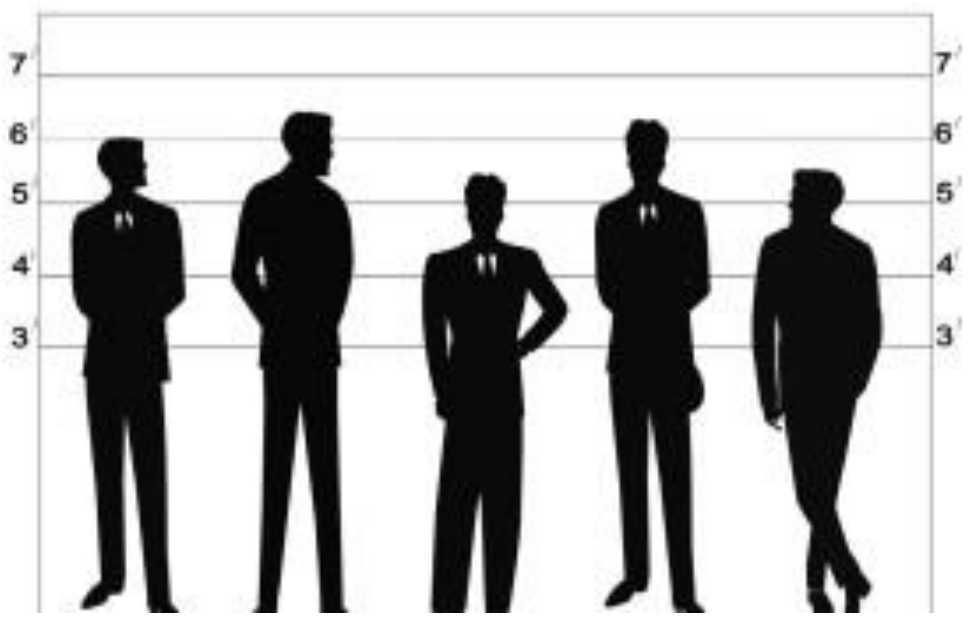
# Which of these are Random variables?











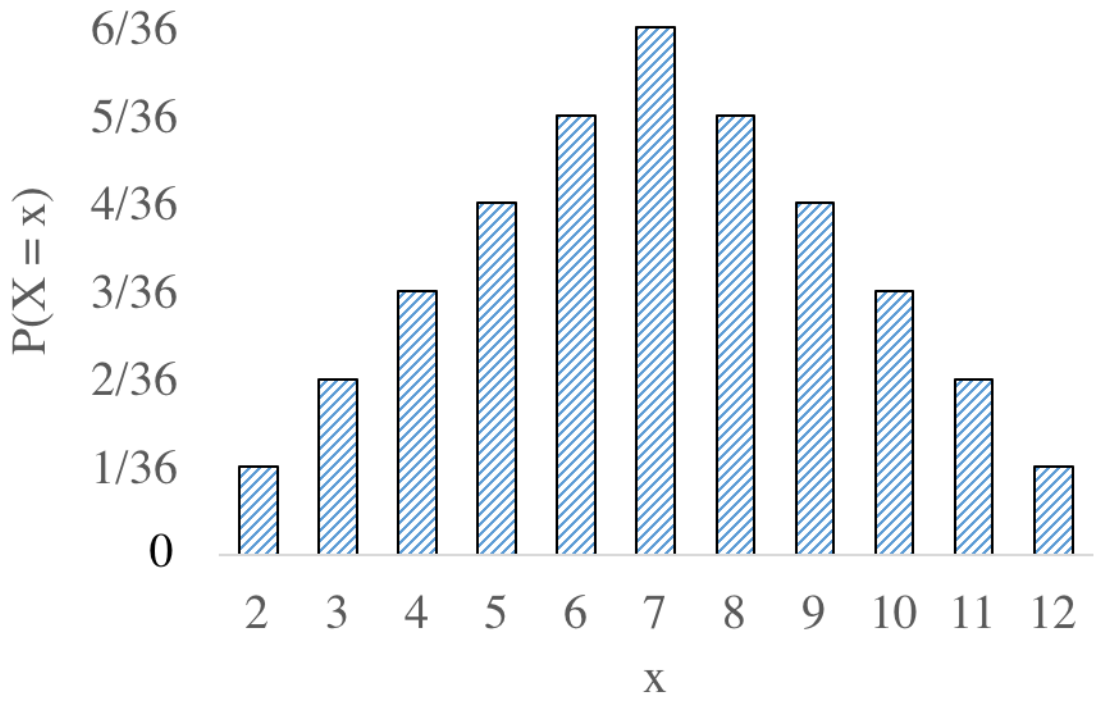
# From empirical frequency to probability distribution

CONSTRUCTION

## Count, then divide

The **probability mass function** (hàm khối xác suất), written p(x), lists every value the variable can take together with the probability attached to each.

- Step 1 — count the occurrences of each value
- Step 2 — divide each count by the total
- Two conditions: p(x) ≥ 0, and the values sum to 1



EXPECTATION AND VARIANCE

# Two investment alternatives

Return over the coming year

| Alternative | Outcomes | Probabilities |
|---|---|---|
| Share A — established issuer | +5% · +15% | 0.5 · 0.5 |
| Share B — growth issuer | −20% · +30% | 0.4 · 0.6 |
| Both alternatives have an expected return of 10% | | |

**Which alternative carries the higher expected return?**

EXPECTATION AND VARIANCE

# Expected value of a discrete random variable

E(X) = Σ x · p(x), computed for Share B

| x | p(x) | x · p(x) |
|---|---|---|
| −20% | 0.4 | −8 |
| +30% | 0.6 | +18 |
| Share A, by the same computation | | E(X) = 10% |
| E(X) = 10%. Neither alternative lists 10% among its possible outcomes. | | |

EXPECTATION AND VARIANCE

# Variance and standard deviation of a random variable

$\mathrm{Var}(X) = \Sigma\,(x - E(X))^2 \cdot p(x)$, computed for Share B

| x | x − E(X) | $(x - E(X))^2$ | weighted by p(x) |
|---|---|---|---|
| −20% | −30 | 900 | 360 |
| +30% | +20 | 400 | 240 |
| Var(X) = 600 | SD(X) ≈ 24.5% | | |
| Variance carries squared units and is not interpretable. Standard deviation carries the units of the variable. | | | |

**Calculate the SD(X) for Share A**

EXPECTATION AND VARIANCE

# The two alternatives compared

**10%**

**EXPECTED RETURN, BOTH**

A figure that appears among the possible outcomes of neither alternative.

**5%**

**STANDARD DEVIATION, SHARE A**

Worst listed outcome +5%; probability of a loss zero.

**24.5%**

**STANDARD DEVIATION, SHARE B**

Worst listed outcome −20%; probability of a loss 0.40.

**Two alternatives with an identical expected value can differ in every other respect that a decision depends on.**

# Expectation and variance in financial decision-making

MECHANISM

## Two summaries of different kinds

A distribution is summarised by two quantities that do not substitute for one another. The expected value locates its centre; the variance measures its spread.

CONSEQUENCES

## What follows for a reported figure

Expected return is never reported without a measure of volatility. The pair Return – Risk is the object of an investment decision.

- An expected value may be a number that cannot occur
- A probability of loss of 0.40 is consistent with a positive expected return
- Session 2 weighted each observation by **1/n**. Session 5 weights each outcome by **p(x)**. The same computation under different weights

# Linear transformation of a random variable

$$\boldsymbol{E(aX + bY + c) = a \cdot E(X) + b \cdot E(Y) + c}$$

**RULES**

**Homogeneity:**

$$E(aX) = a \cdot E(X)$$

$$E(X + c) = E(X) + c$$

**Additivity:**

$$E(X + Y) = E(X) + E(Y)$$

**WHAT IS NOT ADDITIVE?**

**Non-linear function and transformation is not Additive anymore**

$E(X^2) \neq [E(X)]^2$ Unless SD = 0

$E(X \cdot Y) \neq E(X) \cdot E(Y)$ Unless X and Y are totally independent

$$E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)}$$

DISCRETE DISTRIBUTION

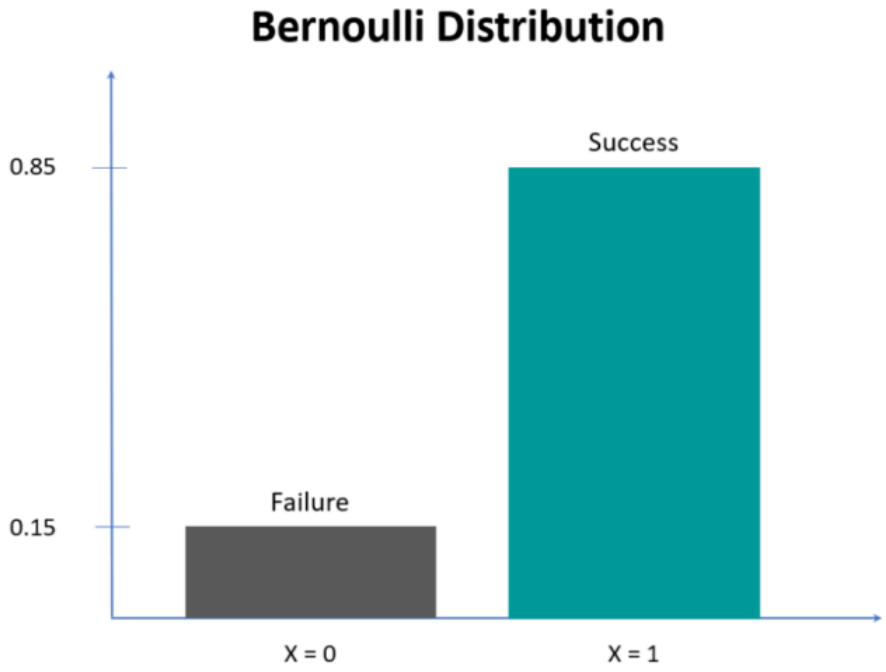
# The Bernoulli Distribution

**ONE TRIAL, TWO OUTCOMES**

## A single parameter p

A **Bernoulli random variable** takes the value 1 with probability **p** and 0 with probability **1 – p**

- E(X) = p
- Var(X) = p(1 – p)
- An outcome that is neither rare nor near-certain carries the greatest uncertainty



DISCRETE DISTRIBUTION

# The Binomial Distribution (Phân bố Nhị thức)

The number of successes in n independent Bernoulli trials

## DEFINITION

**A binomial random variable (biến nhị thức) counts the successes in n independent trials with constant probability p. E(X) = np and Var(X) = np(1 − p)**

## FORMULA

$$P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n-k}$$

**In practice the probability is obtained from software: BINOM.DIST(k, n, p, FALSE) for an exact count.**

## APPLICATION

**Number of defected products**

**Conversion rate in E-commerce platform**

**Medicine effectiveness**

**Guessing answers on a multiple-choice test**

# Assumptions of the Binomial model

**ASSUMPTIONS**

## Three conditions simultaneously

- The number of trials **n** is fixed in advance
- The probability **p** is constant across trials
- The trials are **independent**

**WHERE THEY FAIL**

Independence is an assumption requiring examination in each application, not a default.

- Firms in one industry face a common economic cycle, and in a difficult year the whole industry withholds together
- People are generally affected by ideas of the surrounding
- In real life, the probability may not be fixed due to environment changes

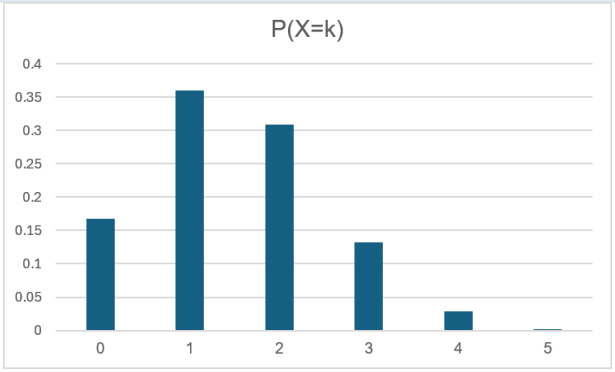
DISCRETE DISTRIBUTION

# Binomial distributions for increasing n

p held constant; the shape of the distribution changes with n alone

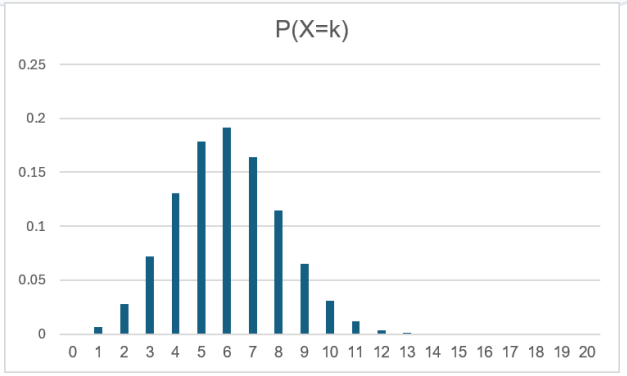
## n = 5

A small number of separated bars, with no discernible outline



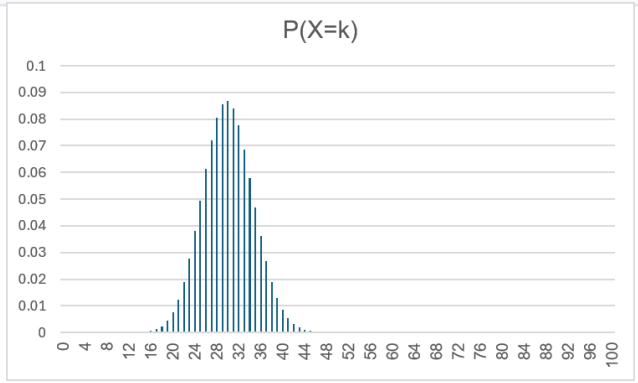

## n = 20

The bars form a symmetric outline around the expected value




## n = 100

The outline approximates a smooth bell-shaped curve




**The convergence of a discrete distribution towards this shape is not explained in this session. It is the subject of the next class**

# The probability of an exact value for a continuous variable

**What is the probability that a student drawn at random has the weight of exactly 50.0000 kg?**

**P(X = x) = 0 for every value x of a continuous random variable.**

A probability of zero does not signify impossibility.

Every value a continuous variable actually takes had probability zero before it was observed.

**Try to apply the chain of thought when working with discrete distribution into this case**

CONTINUOUS DISTRIBUTION: FROM BARS TO AREA

# Narrowing the histogram bin

The same data, drawn with successively narrower bins

| Bin width / Number of Bins | Height of one bar, as a proportion of firms | Total area |
|---|---|---|
| 5 percentage points / 10 Bins | ___ | 1 |
| 2 percentage points / 25 Bins | ___ | 1 |
| 0.5 percentage points / 100 Bins | ___ | 1 |
| 0.1 percentage points / 500 Bins | ___ | 1 |
| Probability is not lost in this limit. It is relocated: from the height of a bar to the area under a curve. | | |

CONTINUOUS DISTRIBUTION: FROM BARS TO AREA

# Probability Density Function (PDF)

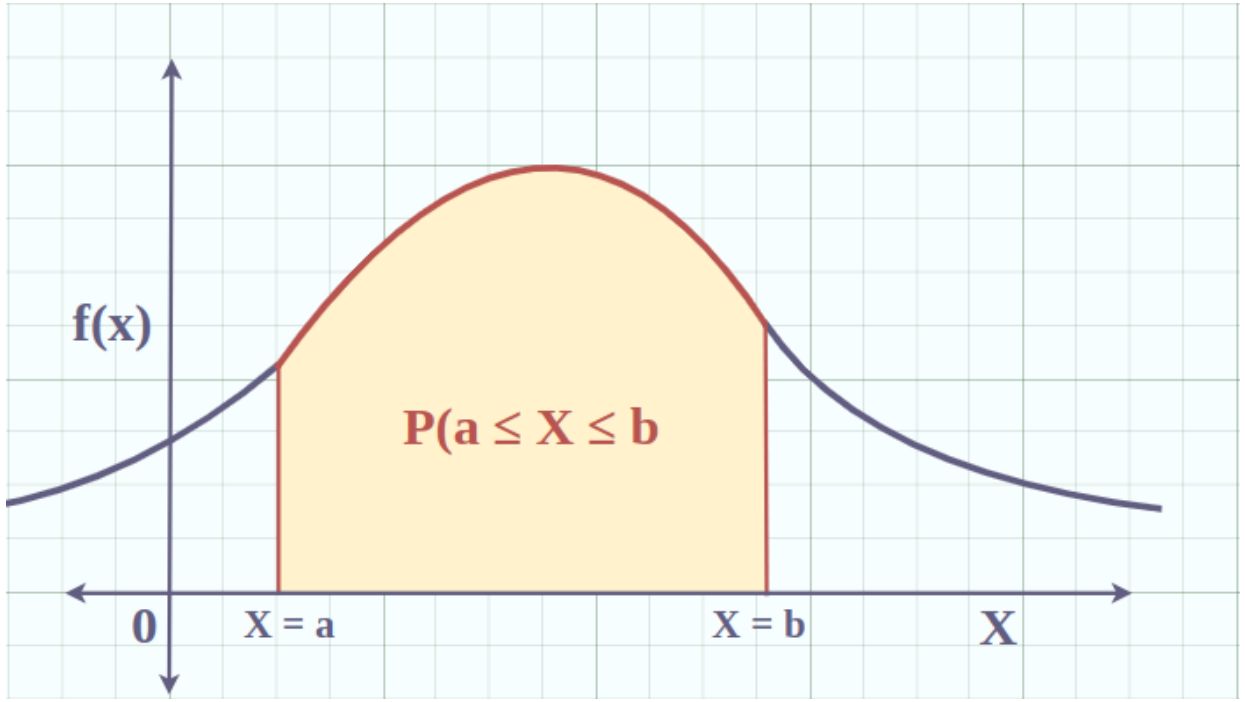
THE VERTICAL AXIS CHANGES MEANING

## Density is proportion per unit of the variable

For a continuous variable the vertical axis records a **probability density** (mật độ xác suất), written **f(x)**, whose unit is proportion per unit of the variable.

- A density is therefore not a probability, and **f(x)** may exceed 1
- Formula for density:

$$P(x \le X \le x + dx) \approx f(x) \cdot dx$$




**DENSITY ≠ PROBABILITY**

CONTINUOUS DISTRIBUTION: FROM BARS TO AREA

# Admissible probability statements for a continuous variable

Three forms of statement are meaningful, and each denotes an area under f(x)

**1**

**P(X > c)**

The area to the right of a threshold.

**2**

**P(X < c)**

The area to the left of a threshold.

**3**

**P(a < X < b)**

The area between two thresholds.

**P(X ≥ c) = P(X > c), since the boundary point contributes zero probability.**

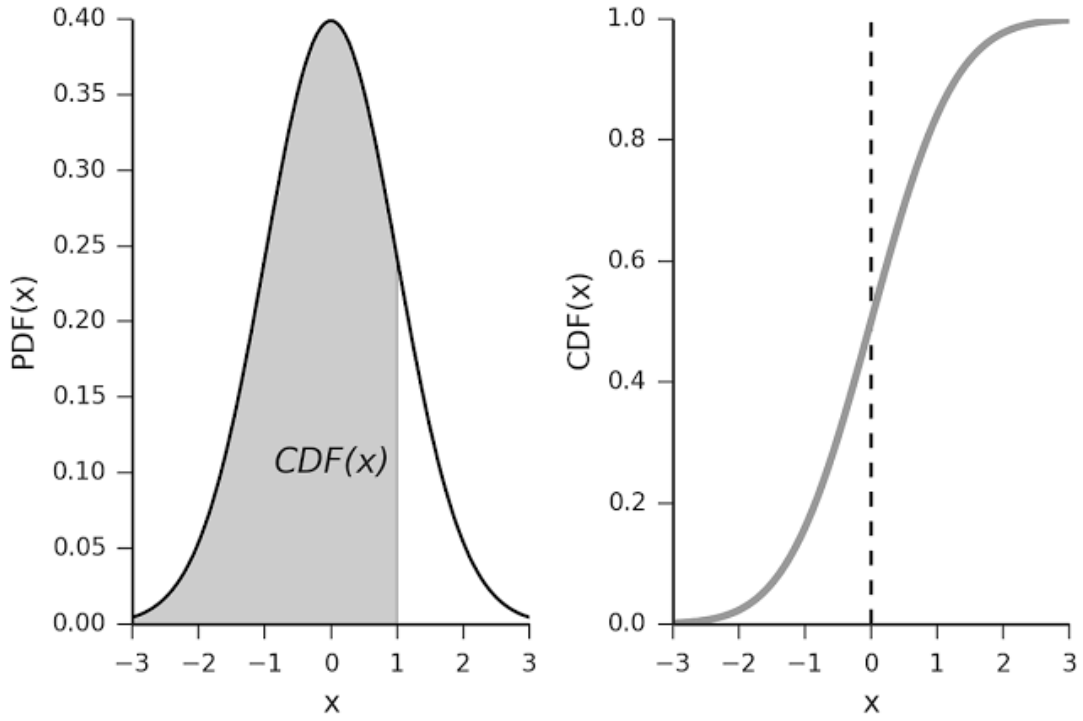
# The Cumulative Distribution Function (CDF)

**ACCUMULATED AREA**

## $F(x) = P(X \leq x)$

The **cumulative distribution function** (hàm phân phối tích lũy) records the area under the density to the left of x. It is non-decreasing, approaching 0 at the lower extreme and 1 at the upper.

- $P(X > c) = 1 - F(c)$
- $P(a < X < b) = F(b) - F(a)$
- F is the function statistical software returns



# Density and cumulative probability in Excel

| Syntax | Returns | Appropriate use |
|---|---|---|
| `NORM.DIST(x, μ, σ, FALSE)` | the density `f(x)` | plotting the curve only |
| `NORM.DIST(x, μ, σ, TRUE)` | the cumulative probability `F(x)` | every probability statement |
| `NORM.S.DIST(z, TRUE)` | `F(z)` for the standard normal | after standardization |
| `NORM.S.INV(p)` | the value `z` such that `F(z) = p` | the inverse; used in Q-Q construction |
| An incorrect final argument produces no error message and returns a well-formed meaningless number. | | |

SUMMARY

# KEY TAKEAWAYS

### 1. RANDOM VARIABLE

An object representing the result of a Random event. The uncertainty is the key to define randomness

### 2. EXPECTED VALUE

An expected value may be a number that cannot occur. It is the centre of the probability mass, not a forecast.

### 3. CONTINUOUS DISTRIBUTION

Every probability statement about a continuous variable is a statement about an area and density

### 4. ZERO

A probability of zero does not signify impossibility. For a continuous variable every exact value has probability zero.

### 5. DISCRETE DISTRIBUTION

Each value of the variable in the Discrete Distribution has its own probability